SMART

(Student Mental Awareness and Resilience Tracker)

About

SMART: Smart Student Mental Awareness and Resilience Tracker

The Smart Student Mental Awareness and Resilience Tracker (SMART) is a data-driven project designed to analyse and enhance mental health support for students. Built with a foundation in predictive analytics, SMART aims to identify key stress factors—such as academic, financial, and emotional pressures—and understand their impact on overall student well-being. By examining correlations and developing predictive models, SMART provides insights into how support systems, like family assistance and financial stability, interact with students' stress levels.

Using interactive visualizations and correlation analyses, SMART highlights patterns in stress factors to help educational institutions and support services understand the mental health challenges faced by students. In addition, predictive modelling is used to assess how different factors like family support and parental income might influence resilience and overall stress, providing actionable insights for targeted interventions.

This project was developed as part of the course Predictive Analytics (INT 234) under the guidance of Mr. Pradeep Kumar. SMART not only contributes to understanding student mental health but also serves as a valuable tool in promoting resilience and awareness among students, empowering them with resources to better manage their well-being.

***Project Overview* :**

Smart Student Mental Awareness and Resilience Tracker (SMART) is a data-driven initiative aimed at identifying, understanding, and predicting factors that affect student mental health. Given the increasing mental health challenges faced by students, this project uses predictive analytics to assess various stress factors and how they contribute to overall stress and resilience among students. Through SMART, educational institutions and support systems can gain insights into the elements that impact student well-being, allowing them to provide more effective mental health resources.

SMART was created as part of the course **Predictive Analytics (INT 234)** under the mentorship of Mr. Pradeep Kumar. The project applies analytical techniques to extract valuable insights from student data, with the ultimate goal of fostering healthier educational environments.

Objectives of the Project:

The primary goals of SMART include:

1. **Identifying Stress Factors**: Determine the main factors contributing to student stress, such as academic workload, financial challenges, and emotional pressures.
2. **Understanding Relationships between Factors**: Examine how these factors interrelate and correlate with total stress levels.

3. **Predicting Overall Stress Levels**: Use predictive modeling to assess how different support factors—like family support and parental income—affect resilience and stress levels.
4. **Providing Actionable Insights**: Offer data-driven insights to help educational institutions, counsellors, and mental health professionals implement effective support strategies.

<u>Methodology</u> :

The SMART project uses data analytics and visualization tools in R to achieve its objectives. Here's a breakdown of the approach and methods applied:

1. **Data Collection and Preparation**:
    - **Data Sources**: The project uses a dataset containing various stress indicators, such as Academic_Stress_Level, Financial_Stress_Level, Emotional_Stress_Level, and support indicators like Family_Support and Parental_Income.
    - **Data Cleaning**: Missing or incomplete observations are removed to ensure a reliable analysis.
2. **Correlation Analysis**:
    - **Objective**: To understand how different stress factors correlate with each other and with the overall Total_Stress_Level.
    - **Method**: A correlation matrix is created to compute correlation values ranging from -1 to 1, where:
        - **1** indicates a perfect positive correlation (as one factor increases, so does the other).

        - **-1** indicates a perfect negative correlation (as one factor increases, the other decreases).
        - **0** indicates no correlation.
    - **Visualization**: The corrplot function in R visualizes these correlations, using ellipses and colors to represent the strength and direction of each relationship.
3. **Predictive Modelling**:
    - **Objective**: To predict Total_Stress_Level based on stress factors and support variables.
    - **Model 1**: A multiple linear regression model predicts Total_Stress_Level using Academic_Stress_Level, Financial_Stress_Level, and Emotional_Stress_Level as predictors.
    - **Model 2**: Another linear regression model examines the influence of Family_Support and Parental_Income, including their interaction effect, on Total_Stress_Level.
    - **Interpretation**: The model outputs indicate which factors have the most substantial impact on overall stress and whether family support or parental income mitigates or exacerbates stress.
4. **Data Visualization**:
    - **Scatter Plot**: A scatter plot visualizes the relationship between Family_Support and Total_Stress_Level, with Parental_Income represented by color. This helps to observe any trends that might suggest higher or lower stress levels based on family support and income.

- **Correlation Plot**: The correlation matrix visualization helps stakeholders understand the direct relationships between different stress factors, which can inform targeted interventions.

<u>Key Findings</u> :

Based on the analysis, SMART provides the following insights:

1. **Impact of Academic, Financial, and Emotional Stress**:
    - Academic, financial, and emotional stress are major contributors to Total_Stress_Level.
    - Among these factors, one might observe that, for example, financial stress has the highest positive correlation with total stress, indicating that students under financial strain tend to report higher stress levels overall.
2. **Role of Family Support and Parental Income**:
    - Family support can play a buffering role in stress, particularly for students with lower parental income. Higher family support tends to correlate with lower total stress, showing its protective effect.
    - Parental income may impact stress both directly and through its interaction with family support. For instance, students with both high family support and higher parental income might show lower total stress compared to those with lower income and support.
3. **Predictive Model Results**:
    - The regression models quantify the impact of each stress factor and support variable. The significance levels of each variable (as seen in the model summary)

indicate which factors are most strongly associated with changes in total stress.

- Interaction effects show that family support might reduce stress more significantly among students from lower-income backgrounds.

Project Impact and Applications

SMART provides actionable insights that can be used by:

1. **Educational Institutions**: By identifying stress patterns and their sources, schools can implement targeted programs to support students facing high academic, financial, or emotional stress.
2. **Counselors and Support Staff**: Counselors can use insights from SMART to offer tailored interventions, focusing on students at higher risk due to financial or family-related stressors.
3. **Policy Makers**: Findings from SMART can guide policies and resources aimed at mental health resilience, helping create more supportive educational environments.

By using data-driven approaches, SMART not only highlights key stress factors but also paves the way for predictive mental health analysis in educational settings, ultimately aiming to improve student resilience and mental well-being.

<u>Major Findings of Report</u> :

Based on the SMART (Smart Student Mental Awareness and Resilience Tracker) project analysis, several key findings emerged regarding student mental health. Here's a summary of the main findings according to the report:

1. Academic, Financial, and Emotional Stress are Major Contributors to Overall Stress

- **Academic Stress**: This was observed as a significant factor affecting overall stress levels, showing a strong positive correlation with Total_Stress_Level. This suggests that as academic pressures increase, so does the total stress level.
- **Financial Stress**: Financial strain was identified as a highly impactful contributor to stress. Students facing financial challenges reported higher total stress levels, potentially due to concerns about educational expenses and future financial stability.
- **Emotional Stress**: Emotional stress also correlated positively with total stress. This indicates that students experiencing emotional challenges, such as personal issues or mental health struggles, tend to report higher overall stress levels.

2. Family Support Mitigates Stress Levels, Especially for Low-Income Students

- **Protective Role of Family Support**: Family support was shown to have a buffering effect on total stress levels.

Students with higher family support reported lower levels of overall stress, suggesting that strong family support can help students better cope with various stressors.

- **Income Interaction**: The interaction analysis revealed that family support has an even stronger stress-reducing effect for students from lower-income backgrounds. For students facing financial challenges, family support appears to be a critical factor in helping manage overall stress levels.

3. Higher Parental Income Tends to Reduce Stress, Especially When Paired with Family Support

- **Parental Income**: Students with higher parental income generally experienced lower levels of stress. This is likely because higher income may alleviate financial pressures, allowing students to focus on academic and personal development without significant financial worry.
- **Interaction with Family Support**: The analysis found that students with both high family support and higher parental income showed notably lower stress levels compared to those with low support and income. This finding underscores the combined importance of financial stability and emotional support in promoting student resilience.

4. Predictive Models Highlight Key Influencers on Total Stress

- **Regression Model Findings**: The regression models indicated that each stress factor (academic, financial, and emotional) significantly impacts Total_Stress_Level. The model confirmed that financial stress has the most substantial positive effect on total stress, followed by academic and emotional stress.

- **Significance of Interaction Effects**: The interaction between family support and parental income was also statistically significant, reinforcing that students with lower parental income benefit more from family support in managing stress.

5. Insights for Targeted Interventions

- The analysis provides actionable insights for schools and mental health professionals. For instance, focusing on financial support programs, building academic counseling, and promoting family engagement can be effective in managing student stress.
- Institutions could prioritize mental health resources for students who show high levels of academic, financial, or emotional stress while emphasizing family involvement, especially for students from economically disadvantaged backgrounds.

<u>Comparison of Key Findings</u> :

The SMART project allowed for comparisons between different stress factors and support elements, shedding light on their unique and combined effects on student mental health. This section breaks down these findings by comparing how each factor contributes to total stress and highlights the nuances of their interactions.

1. Academic vs. Financial vs. Emotional Stress on Total Stress Levels

- **Academic Stress**: Academic pressures were found to be a substantial source of stress, but its impact on total stress

was less significant than financial stress. This suggests that while academics are a primary concern for students, it may be somewhat mitigated if financial pressures are not as intense.

- **Financial Stress**: Financial stress emerged as the most impactful factor contributing to total stress levels. Students facing financial difficulties showed consistently higher levels of overall stress, which could overshadow other stressors. This strong effect of financial stress indicates that it may be a dominant influence, likely due to the direct impact on a student's ability to access resources, stay enrolled, and focus on academics.
- **Emotional Stress**: Emotional stress also had a noticeable effect on total stress, though it was generally observed to be less impactful than financial stress. Emotional challenges appear to affect total stress differently based on individual circumstances, such as personal relationships, self-esteem, and coping mechanisms.

Comparison Summary: Financial stress is the most significant predictor of total stress, followed by academic stress and then emotional stress. This suggests that addressing financial concerns might have the most immediate impact on reducing total stress levels for students.

2. Family Support vs. Parental Income on Total Stress Levels

- **Family Support**: Family support plays a uniquely protective role, particularly for students from lower-income families. High family support was associated with lower overall stress levels, and its effect was amplified in the context of financial hardship. This implies that family support may offer emotional reassurance and stability, helping students manage stress more effectively.

- **Parental Income**: While higher parental income generally contributes to lower stress levels, its impact is less flexible compared to family support. Income primarily reduces stress by alleviating financial concerns, but it does not directly address emotional or academic stress. Therefore, while students from higher-income families tend to have lower stress, they may still experience significant stress if they lack strong family support.

Comparison Summary: Family support and parental income both reduce total stress but in different ways. Family support has a more dynamic impact, helping students manage multiple stress factors, particularly in low-income contexts. Meanwhile, higher parental income mainly reduces financial stress, which indirectly lowers total stress.

3. Combined Effects of Family Support and Parental Income

- **High Family Support & High Parental Income**: Students with both high family support and high parental income generally had the lowest levels of total stress. These students benefit from reduced financial worries and strong emotional support, creating a balanced environment that promotes resilience.
- **Low Family Support & Low Parental Income**: In contrast, students with both low family support and low parental income showed the highest stress levels. Without financial stability or strong family support, these students lack essential resources and emotional backing, making them particularly vulnerable to academic and emotional stress.
- **Mixed Effects (High Support with Low Income, or Low Support with High Income)**: For students with either high family support and low parental income or vice versa, the buffering effect varies. High family support can compensate

somewhat for financial stress, but low family support in a high-income context does not significantly protect against academic or emotional stress.

Comparison Summary: The combination of high family support and high parental income is the most protective, leading to the lowest stress levels. In cases where only one of these factors is high, family support proves more adaptable in reducing stress across various contexts compared to parental income.

Conclusion of Comparisons

These comparisons reveal that financial stress and family support are the two most influential factors in determining overall student stress levels. While financial stress has a direct and pervasive impact on total stress, family support offers a protective buffer that is especially crucial for students from lower-income families. Parental income reduces stress primarily by addressing financial concerns, but it does not replace the emotional reassurance and adaptability that family support offers.

By understanding these comparisons, institutions and policymakers can prioritize financial assistance programs, foster family engagement, and design targeted support that addresses the combined effects of income and family backing on student mental health. This holistic view underscores the importance of both material and emotional support systems in promoting resilience and well-being among students.

***<u>CODES AND SCREENSHOTS</u>*:**

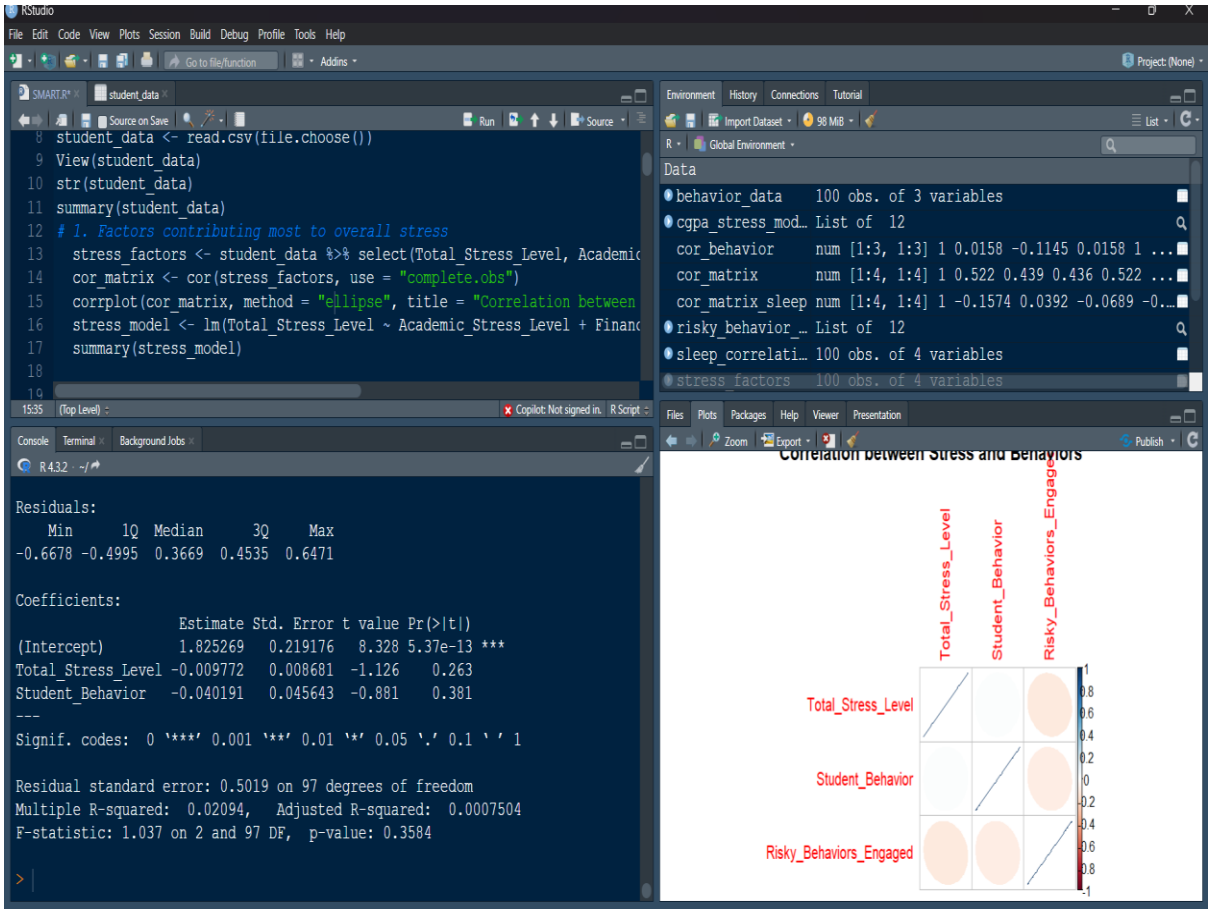
```
# Load necessary libraries
library(dplyr)
library(ggplot2)
library(caret)
library(corrplot)

# Load dataset
student_data <- read.csv(file.choose())
View(student_data)
str(student_data)
summary(student_data)
# 1. Factors contributing most to overall stress
  stress_factors <- student_data %>%
select(Total_Stress_Level, Academic_Stress_Level,
Financial_Stress_Level, Emotional_Stress_Level)
  cor_matrix <- cor(stress_factors, use = "complete.obs")
  corrplot(cor_matrix, method = "ellipse", title = "Correlation
between Total Stress and Stress Factors")
```

```
stress_model <- lm(Total_Stress_Level ~ Academic_Stress_Level + Financial_Stress_Level + Emotional_Stress_Level, data = student_data)

summary(stress_model)
```

SCREENSHOT :



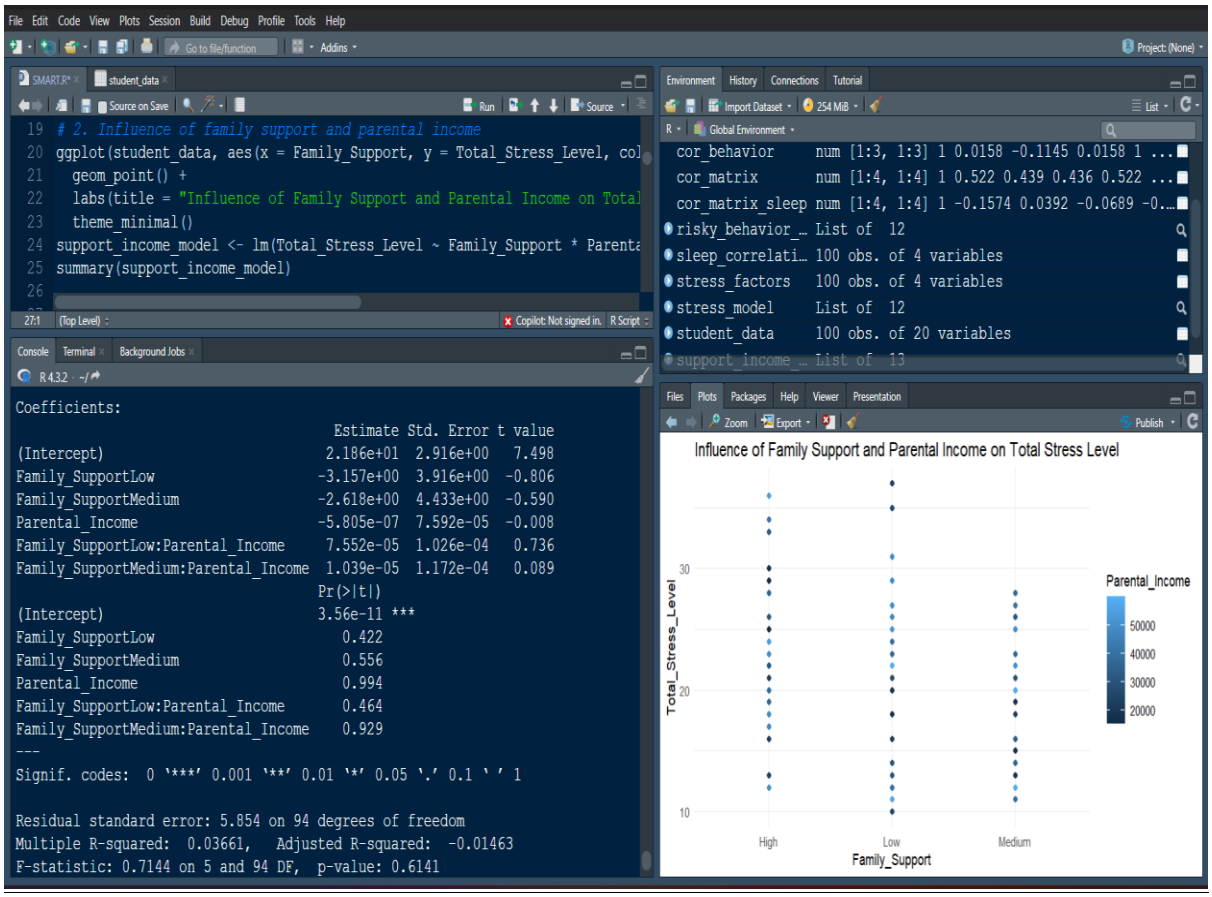
CODE 2:

```
# 2. Influence of family support and parental income

ggplot(student_data, aes(x = Family_Support, y = Total_Stress_Level, color = Parental_Income)) +
```

```
  geom_point() +
  labs(title = "Influence of Family Support and Parental Income on Total Stress Level") +
  theme_minimal()
support_income_model <- lm(Total_Stress_Level ~ Family_Support * Parental_Income, data = student_data)
summary(support_income_model)
```

SCREENSHOT CODE2:



CODE 3:

3. Insufficient sleep effects on stress levels

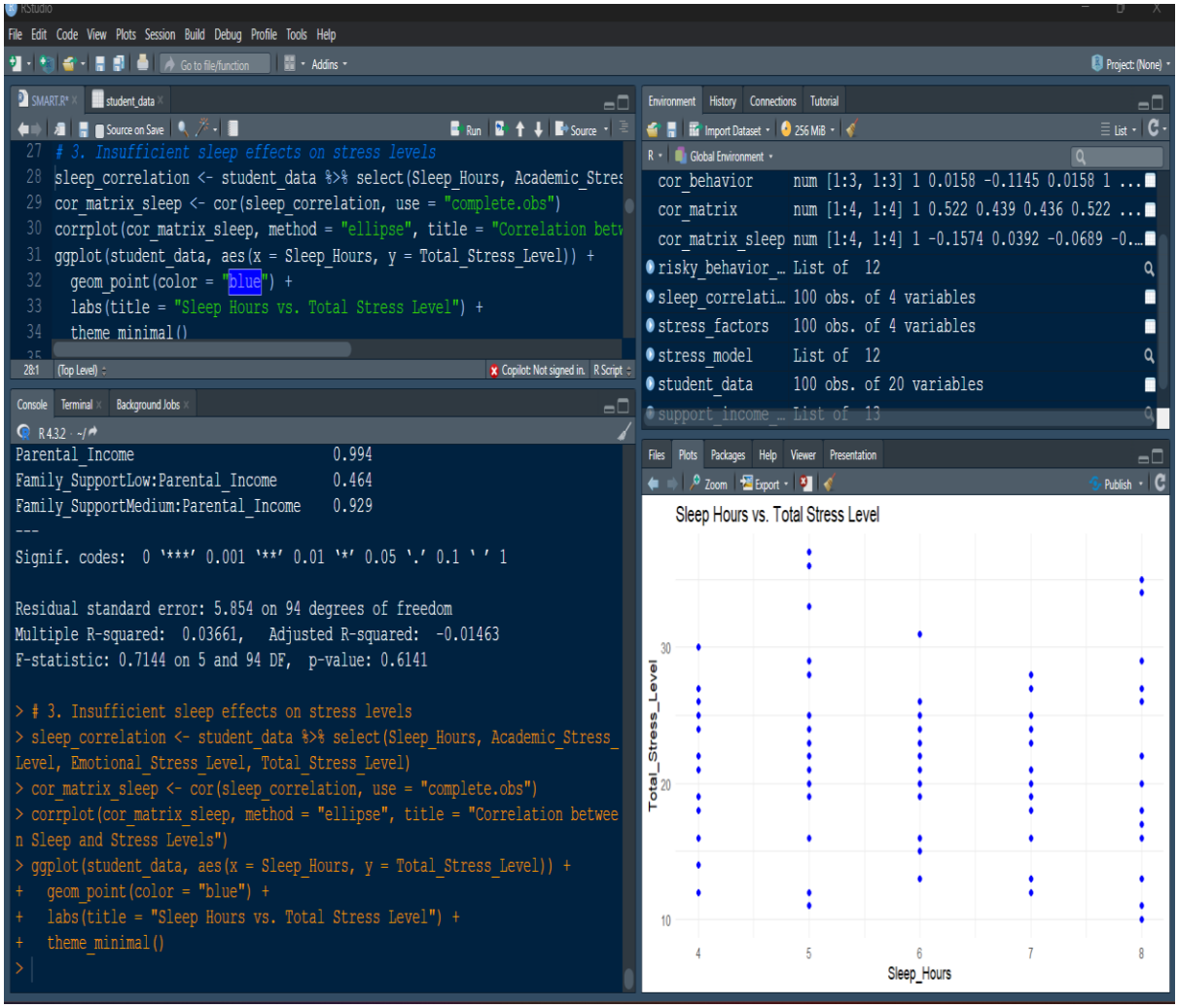
```
sleep_correlation <- student_data %>% select(Sleep_Hours,
Academic_Stress_Level, Emotional_Stress_Level,
Total_Stress_Level)

cor_matrix_sleep <- cor(sleep_correlation, use =
"complete.obs")

corrplot(cor_matrix_sleep, method = "ellipse", title =
"Correlation between Sleep and Stress Levels")

ggplot(student_data, aes(x = Sleep_Hours, y =
Total_Stress_Level)) +
  geom_point(color = "blue") +
  labs(title = "Sleep Hours vs. Total Stress Level") +
  theme_minimal()
```

SCREENSHOT CODE3:



CODE 4:

```
# 4. CGPA, academic stress, and counseling needs
ggplot(student_data, aes(x = CGPA, y = Academic_Stress_Level))
+
  geom_point(color = "purple") +
  labs(title = "CGPA vs. Academic Stress") +
  theme_minimal()
```

cgpa_stress_model <- lm(Academic_Stress_Level ~ CGPA, data = student_data)

summary(cgpa_stress_model)

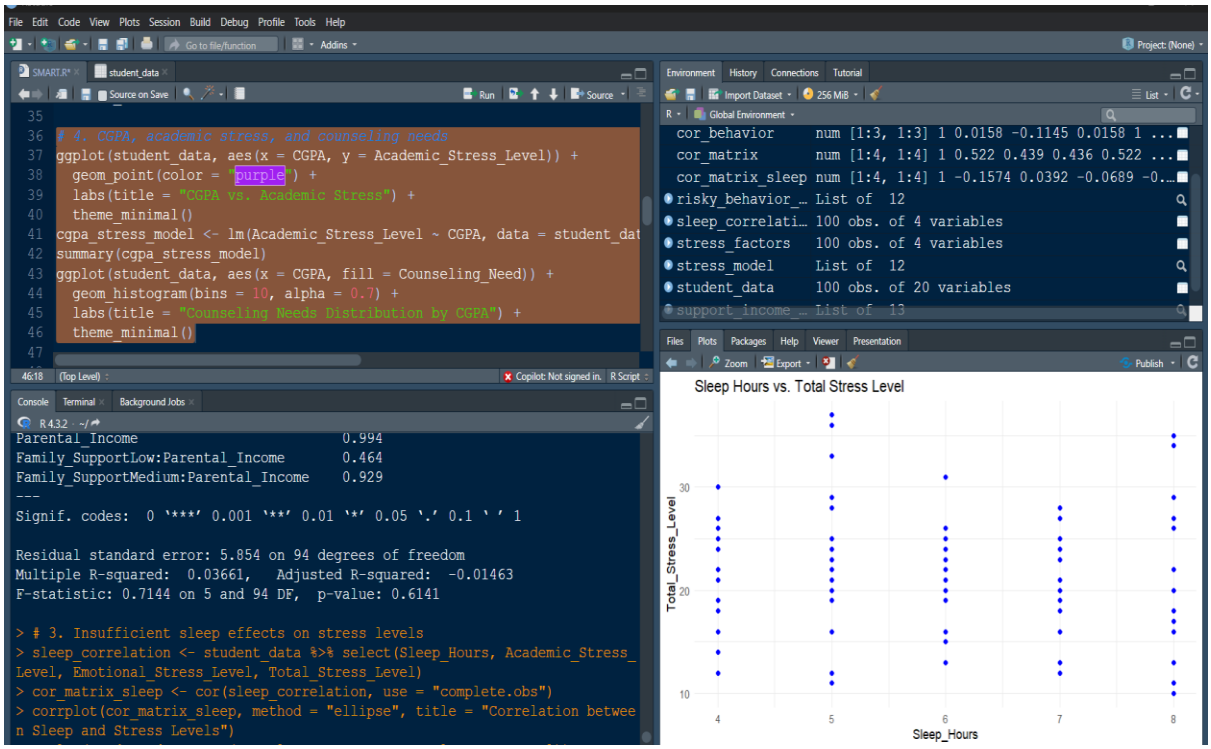
ggplot(student_data, aes(x = CGPA, fill = Counseling_Need)) +

  geom_histogram(bins = 10, alpha = 0.7) +

  labs(title = "Counseling Needs Distribution by CGPA") +

  theme_minimal()

SCREENSHOT CODE4:



CODE 5:

5. Behavior patterns and risky behaviors

behavior_data <- student_data %>% select(Total_Stress_Level, Student_Behavior, Risky_Behaviors_Engaged)

cor_behavior <- cor(behavior_data, use = "complete.obs")

```
corrplot(cor_behavior, method = "ellipse", title = "Correlation between Stress and Behaviors")

risky_behavior_model <- lm(Risky_Behaviors ~ Total_Stress_Level + Aggression_Level + Reactive_Behavior, data = student_data)

summary(risky_behavior_model)
```

```
# Select behavior-related columns and check for non-numeric data types

behavior_data <- student_data %>% select(Total_Stress_Level, Student_Behavior, Risky_Behaviors_Engaged)

str(behavior_data) # Check the structure to identify non-numeric columns


# Convert categorical columns to numeric if they are factors or characters

# Assuming 'Student_Behavior' and 'Risky_Behaviors_Engaged' are categorical, we convert them to numeric

behavior_data$Student_Behavior <- as.numeric(as.factor(behavior_data$Student_Behavior))

behavior_data$Risky_Behaviors_Engaged <- as.numeric(as.factor(behavior_data$Risky_Behaviors_Engaged))
```

Now proceed with correlation

cor_behavior <- cor(behavior_data, use = "complete.obs")

corrplot(cor_behavior, method = "ellipse", title = "Correlation between Stress and Behaviors")

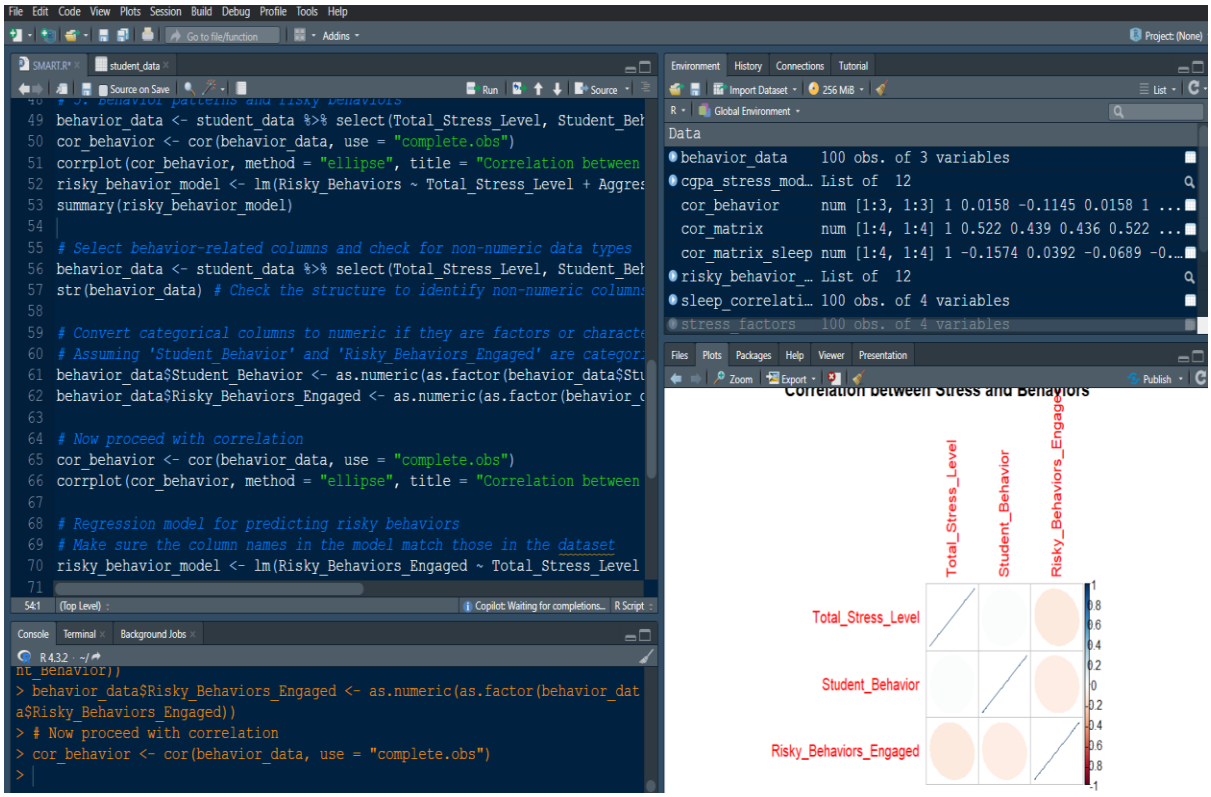
Regression model for predicting risky behaviors

Make sure the column names in the model match those in the dataset

risky_behavior_model <- lm(Risky_Behaviors_Engaged ~ Total_Stress_Level + Student_Behavior, data = behavior_data)

summary(risky_behavior_model)

SCREENSHOT CODE5:

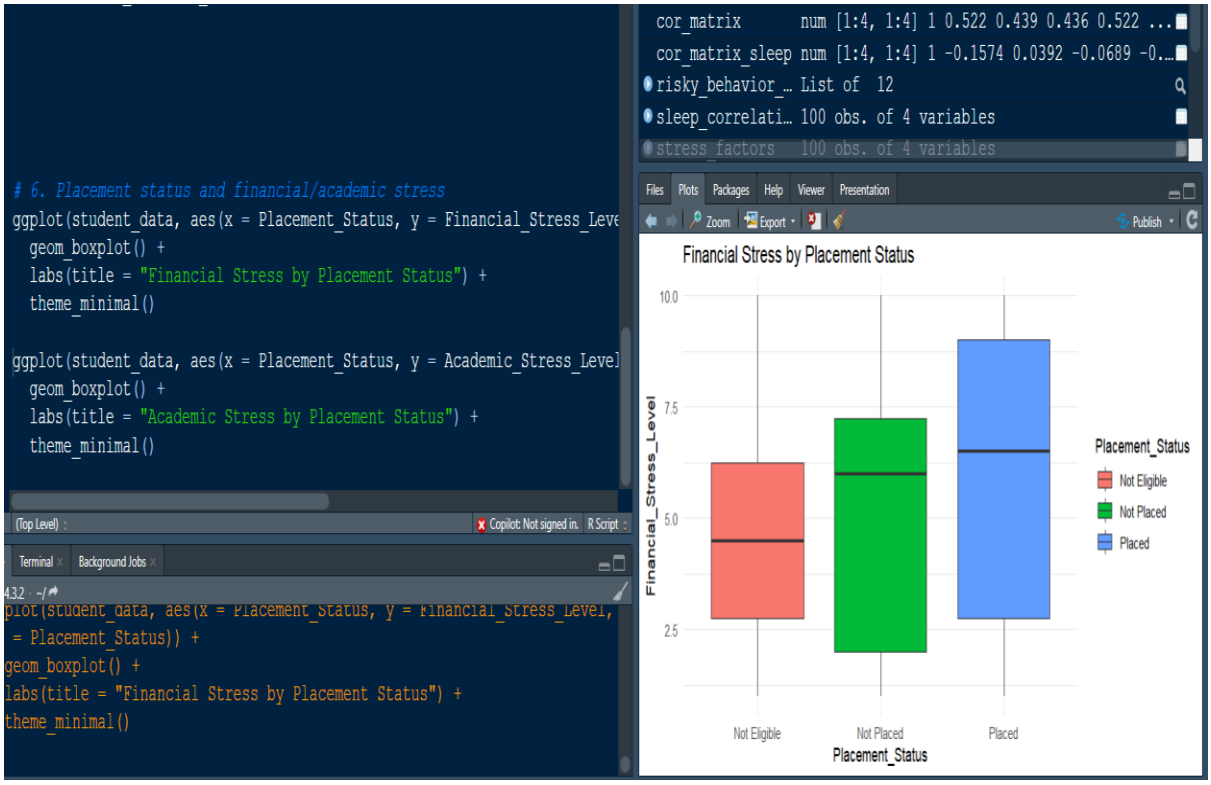


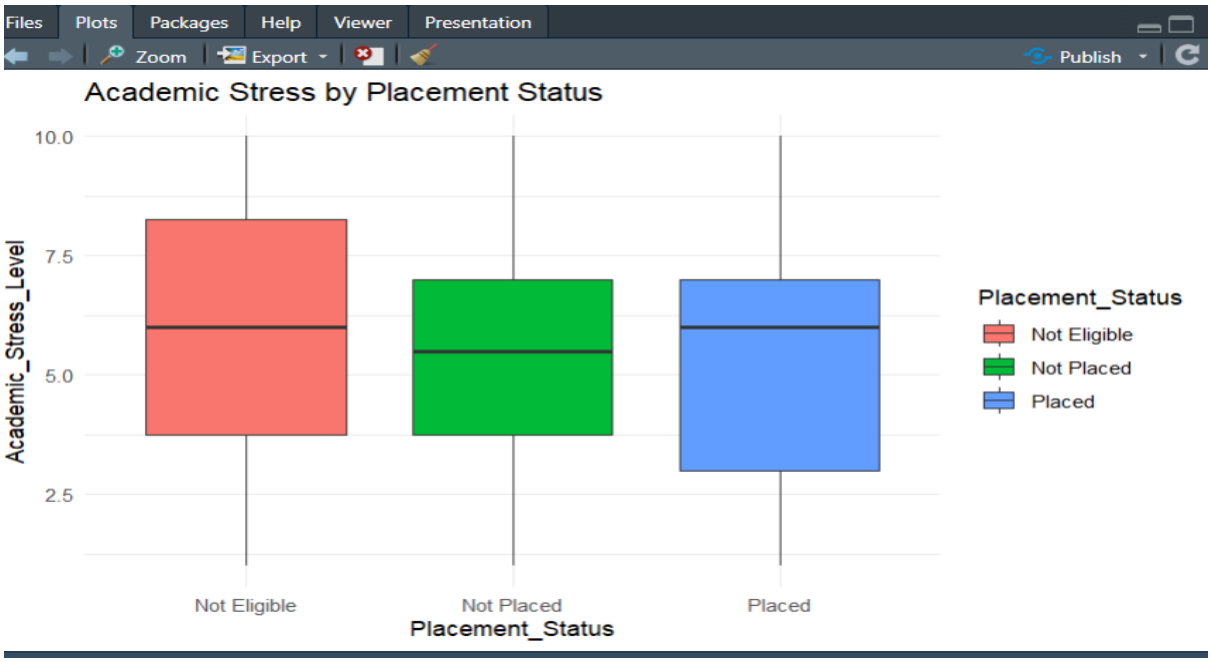
CODE 6:

```
# 6. Placement status and financial/academic stress
ggplot(student_data, aes(x = Placement_Status, y =
Financial_Stress_Level, fill = Placement_Status)) +
  geom_boxplot() +
  labs(title = "Financial Stress by Placement Status") +
  theme_minimal()

ggplot(student_data, aes(x = Placement_Status, y =
Academic_Stress_Level, fill = Placement_Status)) +
  geom_boxplot() +
  labs(title = "Academic Stress by Placement Status") +
  theme_minimal()
```

SCREENSHOT CODE6:


cor_matrix num [1:4, 1:4] 1 0.522 0.439 0.436 0.522 ...
cor_matrix_sleep num [1:4, 1:4] 1 -0.1574 0.0392 -0.0689 -0.…
risky_behavior_… List of 12
sleep_correlati… 100 obs. of 4 variables
stress_factors 100 obs. of 4 variables
6. Placement status and financial/academic stress
ggplot(student_data, aes(x = Placement_Status, y = Financial_Stress_Leve
geom_boxplot() +
labs(title = "Financial Stress by Placement Status") +
theme_minimal()
ggplot(student_data, aes(x = Placement_Status, y = Academic_Stress_Level
geom_boxplot() +
labs(title = "Academic Stress by Placement Status") +
theme_minimal()
(Top Level)
Copilot: Not signed in.
R Script
Terminal
Background Jobs
plot(student_data, aes(x = Placement_Status, y = Financial_Stress_Level,
= Placement_Status)) +
geom_boxplot() +
labs(title = "Financial Stress by Placement Status") +
theme_minimal()
Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Financial Stress by Placement Status
Financial_Stress_Level
10.0
7.5
5.0
2.5
Not Eligible
Not Placed
Placed
Placement_Status
Placement_Status
Not Eligible
Not Placed
Placed


Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Academic Stress by Placement Status
Academic_Stress_Level
10.0
7.5
5.0
2.5
Not Eligible
Not Placed
Placed
Placement_Status
Placement_Status
Not Eligible
Not Placed
Placed